IES

Lab 3.1 · FMEA Document

Build an FMEA and calculate RPN in Excel.

Goal	Build an FMEA and calculate RPN in Excel.
Output	Ranked FMEA with severity gate, actions and evidence gaps.
Time box	Use the Lesson Plan time; stop at the stated module boundary.

Concept and workflow

Frame the engineering question and identify the permitted evidence. Plan the output and acceptance check. Generate with Copilot. Inspect every source, formula, unit and assumption. Verify one result independently. Correct one precise gap, rerun, and save the evidence.

Prompt to use

Use only FMEAInput. Preserve ratings. Calculate RPN=S*O*D; flag Severity >=9.

Engineering boundary

Higher Detection means poorer detectability; no invented ratings.

Evidence to retain

Ranked FMEA with severity gate, actions and evidence gaps. Save the final prompt, source filenames, one independent check, correction history and reviewer.

Microsoft 365 route and checks

Open https://m365.cloud.microsoft/ and choose the relevant Copilot surface. For agents use https://m365.cloud.microsoft/chat/ and the Agents entry. For SharePoint use https://www.microsoft365.com/launch/sharepoint. Interface labels may vary by tenant.

Acceptance checklist

- Correct source and revision selected
- Output matches the requested format
- Formula, unit or citation independently checked
- Unknowns remain visible
- No invented cause, limit or approval
- Human reviewer and release decision recorded